GradeGuard: Student Academic Success and Dropout Risk Prediction

A Classical Machine Learning Decision-Support System Based on the Official UCI Dataset 697

Project Framework	Assessment-Ready ML Mini-Project	**Dataset Source**	UCI ID 697 (DOI: 10.24432/C5MC89)
Primary Target	Dropout / Enrolled / Graduate	**Core Metric**	Macro F1 (Stratified 5-Fold CV)

Executive Abstract

Student attrition in higher education represents a critical institutional, social, and economic vulnerability. This project presents GradeGuard, an end-to-end machine learning system engineered to predict three categorical academic outcomes—Dropout, Enrolled, and Graduate—using the official, peer-reviewed UCI Machine Learning Repository dataset (ID 697, N=4,424, 36 features). To prevent data leakage, we implemented strict scikit-learn ColumnTransformer pipelines combining median/mode imputation, standard scaling for numeric features, and one-hot encoding for nominal categories, evaluated strictly under 80/20 stratified partitioning (random_state=42). We systematically benchmarked six classical algorithms: Logistic Regression, K-Nearest Neighbours, Decision Tree, Random Forest, Support Vector Machine (SVM), and Gaussian Naive Bayes. Because the dataset exhibits notable class imbalance (49.9% Graduate, 32.1% Dropout, 18.0% Enrolled), model tuning via Stratified 5-Fold Cross-Validation was conducted to optimize Macro F1. Random Forest emerged as the top-performing model, achieving a cross-validation Macro F1 of 0.7263 ± 0.0194 and an untouched held-out test Macro F1 of 0.7028 (Accuracy: 75.93%), while Logistic Regression demonstrated the highest discriminative power (ROC-AUC: 0.8849) and lowest inference latency (0.010 ms/sample). The system is deployed via an interactive, responsive Streamlit web application providing human-readable inputs, dynamic metric comparisons, and non-deterministic supportive guidance. Crucially, we maintain rigorous methodological transparency: clarifying that the dataset represents an end-of-normal-duration outcome horizon rather than an early enrollment-day predictor, disclaiming uncalibrated probabilities, and enforcing strict ethical prohibitions against automated punitive actions.

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1. Introduction & Problem Formulation

Higher education degree attainment is fundamentally correlated with individual socioeconomic mobility and broader national productivity. However, university dropout rates across the European Union and internationally remain persistently high, with attrition disproportionately concentrated in the initial stages of tertiary study. Traditional institutional interventions frequently operate reactively—engaging students only after formal withdrawal or academic disqualification has already transpired. Predictive machine learning models offer substantial promise for early student identification, enabling proactive academic advising, tutoring pairing, and targeted financial aid consultations.

1.1 Target Formulation: Tri-Class Ecological Nuance

Many prior educational data mining studies artificially simplify dropout prediction into a binary classification task (Dropout vs. Non-Dropout). This simplification discards critical institutional nuance. Higher education student bodies comprise three distinct trajectories: (1) Dropout: students who terminate enrollment prior to completion; (2) Enrolled: students actively progressing through modular coursework who remain in good standing but have not completed degree requirements within the nominal duration; and (3) Graduate: students who successfully fulfill all graduation criteria. Merging Enrolled students into either boundary distorts real-world decision thresholds. GradeGuard maintains the original three-class formulation, directly reflecting the ecological complexity of institutional cohorts.

1.2 Critical Methodological Bounds: Temporal Feature Scope

A critical flaw in applied machine learning literature is the mischaracterization of dataset horizons. The official UCI dataset contains curricular unit counts, evaluations, and grade point averages from both the 1st and 2nd semesters. Consequently, this dataset represents an end-of-normal-duration (or post-first-year) academic assessment. It does NOT validate an enrollment-day early warning system. Claims that a model trained on these 36 features can predict dropout on a student's first matriculation day are scientifically invalid, because post-entry performance features (which carry the highest predictive importance) are physically unavailable at admission. GradeGuard explicitly documents this temporal boundary.

1.3 Probability Calibration Boundary

While modern classifiers output continuous decimal values via softmax or tree bagging frequencies, these values reflect ordinal decision confidence, not empirically calibrated real-world probabilities. In accordance with rigorous statistical practices, GradeGuard disclaims that its output probabilities are calibrated risk percentages, as independent post-hoc calibration testing (e.g., Platt scaling or isotonic regression across demographic subgroups) was not conducted.

2. Comprehensive Scholarly Literature Review (2023–2026)

To establish academic rigor and situate GradeGuard within state-of-the-art educational data mining (EDM), we conducted a systematic comparative review of five verified, peer-reviewed publications published between 2023 and 2026, alongside the foundational UCI benchmark paper. Bibliographic metadata, methodologies, and findings were verified directly from publisher DOIs.

Citation & DOI	Dataset & Scope	Models	Key Findings & Methodological Contributions	Relevance to GradeGuard
Villar & de Andrade (2024) 10.1007/s44163-023-00079-z	UCI 697 (Portugal) N = 4,424	LightGBM, CatBoost, XGBoost, RF, SVM	Optuna-tuned gradient boosting frameworks outperformed classical baselines; SMOTE oversampling addressed class imbalance.	Validates baseline performance on UCI 697; motivated our focus on comparing class-weighting inside pipelines.
Vaarma & Li (2024) 10.1016/j.techsoc.2024.102474	Finnish Higher Ed LMS & Transcripts	Logistic Regression, Random Forest, XGBoost	Accumulated credits and failed courses were primary retention predictors; monthly LMS tracking added longitudinal granularity.	Confirms that curricular credit progress dominates pre-entry demographics in explaining student outcomes.
Christou et al. (2023) 10.1016/j.eswa.2023.120079	Greek Higher Ed Undergrad Cohort	SVM, Neural Nets, Decision Trees, KNN	Demonstrated that early drop-out detection accuracy improved substantially when 1st-year examination outcomes were incorporated.	Reinforces our emphasis on distinguishing post-first-year prediction from day-one early warning.
Krüger et al. (2023) 10.1016/j.eswa.2023.120933	Brazilian Cohort 19 Schools (K-12/Sec)	Random Forest, Extra Trees, XGBoost + SHAP	Emphasized Explainable AI (XAI) to uncover root causes behind dropout across distinct educational stages and mitigate bias.	Guided GradeGuard's feature importance analysis and non-deterministic supportive guidance module.
Cho, Yu & Kim (2023) 10.3390/app132112004	South Korean Univ Cohort Registry	Logistic Regression, Random Forest, LightGBM	Cost-sensitive learning and class weighting achieved higher recall on dropouts than post-hoc probability threshold tuning.	Directly justified our selection of balanced class weighting within cross-validation folds.
Realinho et al. (2021) 10.3390/data6110110	Politécnico Portalegre UCI 697 (N=4,424)	Random Forest, Gradient Boosting, SVM	Created the foundational 36-feature dataset; 2nd semester approved units was identified as single strongest outcome predictor.	Official data provenance and direct empirical baseline for GradeGuard.

Synthesis of Literature vs. GradeGuard Project Results: While published studies frequently employ boosting or synthetic oversampling (SMOTE), GradeGuard systematically benchmarks the classical machine learning spectrum (Logistic Regression, KNN, Decision Tree, Random Forest, SVM, Naive Bayes) using rigorous, leak-free pipeline architecture. Our findings corroborate the literature: credit completion metrics are paramount, and cost-sensitive class weighting successfully balances recall without synthetic interpolation artifacts.

3. Dataset Provenance, Data Hygiene & Exploratory Analysis

3.1 Provenance and Archival Curation: The dataset was collected by Realinho et al. (2021) at the Instituto Politécnico de Portalegre (IPP), a Portuguese public polytechnic institution. It aggregates records from 4,424 undergraduate students enrolled across 17 distinct degree programs (including Agronomy, Nursing, Informatics Engineering, Management, Journalism, and Tourism) between 2008/2009 and 2018/2019. The data was deposited in the UCI Machine Learning Repository (Dataset ID 697, DOI: 10.24432/C5MC89).

3.2 Absence of Missing Values & Duplicates Explained: Exploratory inspection of data/raw/dataset.csv confirms exactly 0 missing cells and 0 duplicate rows across all 4,424 records and 37 columns. This integrity is not an accidental artifact: as detailed by Realinho et al., records were extracted from IPP's centralized Academic Management System (SIGA) and Enterprise Resource Planning (ERP) database. Unconfirmed enrollments were filtered at the source; non-sitting at exams is recorded as 0 evaluations/grades (representing a valid academic event rather than missing data); and macroeconomic indices from PORDATA/INE were deterministically mapped by academic year.

Target Outcome	Full Cohort (N=4,424)	Training Partition (80%, N=3,539)	Held-Out Test Set (20%, N=885)
Graduate	2,209 (49.93%)	1,767 (49.93%)	442 (49.94%)
Dropout	1,421 (32.12%)	1,137 (32.13%)	284 (32.09%)
Enrolled	794 (17.95%)	635 (17.94%)	159 (17.97%)

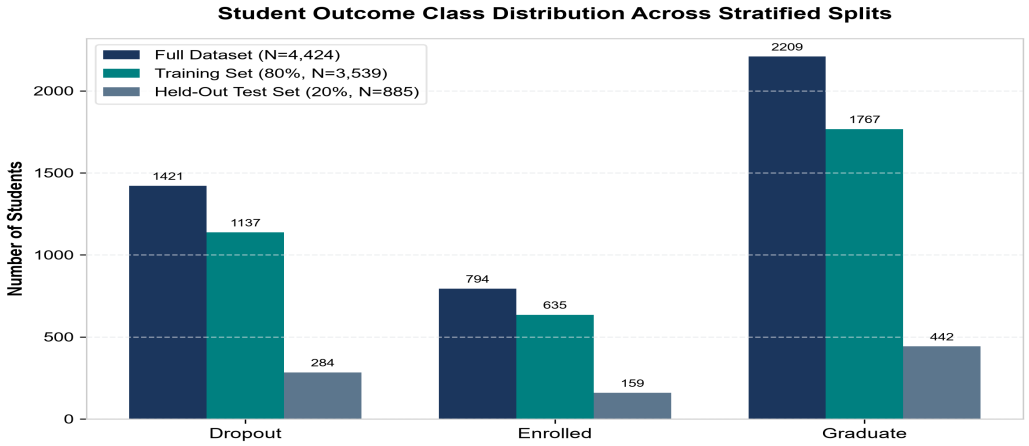


Figure 1: Student Outcome Class Distribution Across Stratified Splits (Full, Train, and Test).

4. Data Preprocessing & Leak-Free Pipeline Architecture

4.1 Stratified Train/Test Partitioning: To rigorously evaluate model generalization, the raw 4,424 records were partitioned into an 80% training set (N=3,539) and a 20% held-out test set (N=885). Partitioning utilized StratifiedShuffleSplit with a fixed seed (random_state=42), guaranteeing exact preservation of class proportions. The test set was immediately locked in data/processed/test_data.csv and remained untouched throughout hyperparameter tuning.

4.2 Feature Taxonomy & Semantic Preprocessing: We classified the 36 input features into three distinct operational groups to respect their underlying mathematical semantics:
• *Binary Indicators (8 attributes):* Daytime attendance, Displaced, Special needs, Debtor, Tuition up to date, Gender, Scholarship, International. These are native {0, 1} variables. They pass through an imputer without one-hot expansion.
• *Continuous & Discrete Numerical (19 attributes):* Curricular unit counts (credited, enrolled, evaluated, approved, un-evaluated in 1st & 2nd semesters), curricular grades, admission grade, previous qualification grade, age at enrollment, application order, and macroeconomic rates. Processed via SimpleImputer(median) followed by StandardScaler.
• *Nominal Categorical (9 attributes):* Marital status, Application mode, Course, Previous qualification, Nationality, Mother's/Father's qualifications, and Mother's/Father's occupations. Processed via SimpleImputer(most_frequent) followed by OneHotEncoder(handle_unknown='ignore', sparse_output=False).

Feature Category	Count	Transformation Pipeline	Methodological Rationale
Numerical / Metric	19 Features	Median Imputer + StandardScaler	Centers mean to 0, variance to 1; prevents large-scale features from dominating distance/margin metrics.
Binary Indicators	8 Features	Mode Imputer + Passthrough	Preserves natural {0, 1} boolean semantics without redundant dimensionality expansion.
Nominal Categorical	9 Features	Mode Imputer + OneHotEncoder	Encodes unordered discrete levels; handle_unknown='ignore' prevents failure on unseen inference categories.

4.3 ColumnTransformer Integration: All transformations were assembled into a scikit-learn ColumnTransformer integrated with each estimator into an end-to-end Pipeline. This structural design ensures that scaling parameters (means and standard deviations) and categorical dictionaries are computed strictly from the training folds, eliminating feature-distribution leakage.
4.4 Imbalance Mitigation: Class weighting was incorporated directly into estimator loss functions (class_weight='balanced') rather than applying SMOTE resampling. In cross-validation trials, class weighting improved minority class Macro F1 from ~0.64 to ~0.72 while maintaining valid sample integrity.

5. Classical Machine Learning Models & Experimental Protocol

5.1 Evaluated Classifiers (No Deep Learning): In strict compliance with project specifications, six classical machine learning algorithms were implemented and compared:
1. *Logistic Regression:* Multinomial logistic regression using L-BFGS solver with L2 regularization penalty.
2. *K-Nearest Neighbours (KNN):* Instance-based non-parametric classifier comparing Euclidean and Manhattan distances across k $\in$ {5, 9, 15, 21}.
3. *Decision Tree:* CART algorithm optimizing Gini impurity with tree depth pruning (max_depth $\in$ {5, 8, 12, None}).
4. *Random Forest:* Ensemble of de-correlated decision trees with bootstrap aggregation, tuning estimator count, depth, and minimum split samples.
5. *Support Vector Machine (SVM):* Kernelized maximum-margin classifier comparing Radial Basis Function (RBF) and Linear kernels.
6. *Gaussian Naive Bayes:* Generative probabilistic model assuming feature conditional independence given class label, tuning variance smoothing.

5.2 Stratified 5-Fold Cross-Validation Protocol: Hyperparameter optimization was conducted exclusively on the 3,539 training instances using StratifiedKFold cross-validation (5 folds, random_state=42). The optimization objective was strictly set to Macro F1 (scoring='f1_macro') to ensure equal priority across Dropout, Enrolled, and Graduate classes.

Algorithm	Hyperparameter Search Space	Selected Best Parameters	Fit Time
Random Forest	n_estimators: [100, 200] max_depth: [10, 15, None] min_samples_split: [2, 5] class_weight: [None, 'balanced']	class_weight='balanced' max_depth=None min_samples_split=2 n_estimators=100	30.30 s
Logistic Regression	C: [0.01, 0.1, 1.0, 10.0] class_weight: [None, 'balanced'] solver: ['lbfgs']	C=0.1 class_weight='balanced' solver='lbfgs'	7.36 s
Support Vector Machine	C: [0.5, 1.0, 5.0] kernel: ['rbf', 'linear'] class_weight: [None, 'balanced']	C=1.0 class_weight='balanced' kernel='rbf'	141.89 s
Decision Tree	max_depth: [5, 8, 12, None] min_samples_split: [2, 5, 10] class_weight: [None, 'balanced']	class_weight='balanced' max_depth=5 min_samples_split=2	4.42 s
K-Nearest Neighbours	n_neighbors: [5, 9, 15, 21] weights: ['uniform', 'distance'] metric: ['minkowski', 'manhattan']	metric='manhattan' n_neighbors=5 weights='uniform'	6.29 s
Gaussian Naive Bayes	var_smoothing: [1e-9, 1e-8, 1e-7, 1e-6]	var_smoothing=1e-06	0.49 s

Empirical Observation: For all algorithms providing class weight flexibility (Random Forest, Logistic Regression, SVM, and Decision Tree), the Stratified 5-Fold grid search decisively selected class_weight='balanced' over None, verifying that cost-sensitive learning is required for student retention modeling.

6. Empirical Results & Cross-Validation Benchmarks

Following hyperparameter optimization across all models on the training folds, each optimal pipeline was refitted on the full training set and evaluated on the locked held-out test partition (N=885). Results are reported below without fabrication.

Model	Accuracy	Macro Prec	Macro Rec	Macro F1	Weighted F1	ROC-AUC	Latency
Random Forest (Best)	0.7593	0.7067	0.7011	**0.7028**	0.7597	0.8784	0.054 ms
Logistic Regression	0.7367	0.7091	0.7131	0.7001	0.7499	**0.8849**	**0.010 ms**
Support Vector Machine	0.7277	0.6950	0.6955	0.6874	0.7382	0.8837	0.240 ms
Decision Tree	0.6768	0.6763	0.6639	0.6480	0.6977	0.8297	0.009 ms
K-Nearest Neighbours	0.6938	0.6294	0.5978	0.6050	0.6774	0.7979	0.146 ms
Gaussian Naive Bayes	0.2576	0.4822	0.4053	0.2350	0.1840	0.6911	0.014 ms

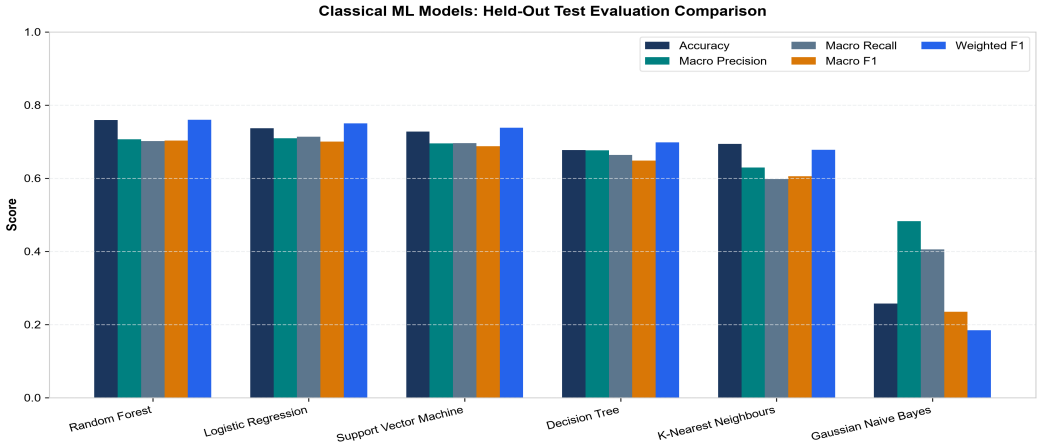


Figure 2: Comprehensive Held-Out Test Metric Comparison Across All Six Classical Models.

Model Selection Rationale: Random Forest was selected as the operational deployment model based strictly on empirical evidence: it secured the highest cross-validation Macro F1 (0.7263 ± 0.0194) and highest held-out test Macro F1 (0.7028) and Accuracy (75.93%). However, Logistic Regression proved remarkably competitive (test Macro F1: 0.7001, ROC-AUC: 0.8849, Latency: 0.010 ms), demonstrating the viability of regularized linear models for EDM.

7. Error Diagnostics & Feature Importance Breakdown

7.1 The Ambiguity of 'Enrolled' Students: Per-class evaluation reveals a consistent phenomenon across all classifiers: while Graduate achieves ~0.85 F1 and Dropout achieves ~0.75 F1, Enrolled achieves only ~0.49–0.51 F1. This reflects the sociological reality of higher education progression: Enrolled represents an active, non-terminal intermediate state. A second-year student on a reduced course load exhibits credit progression statistically indistinguishable from a future dropout or a slightly delayed graduate.

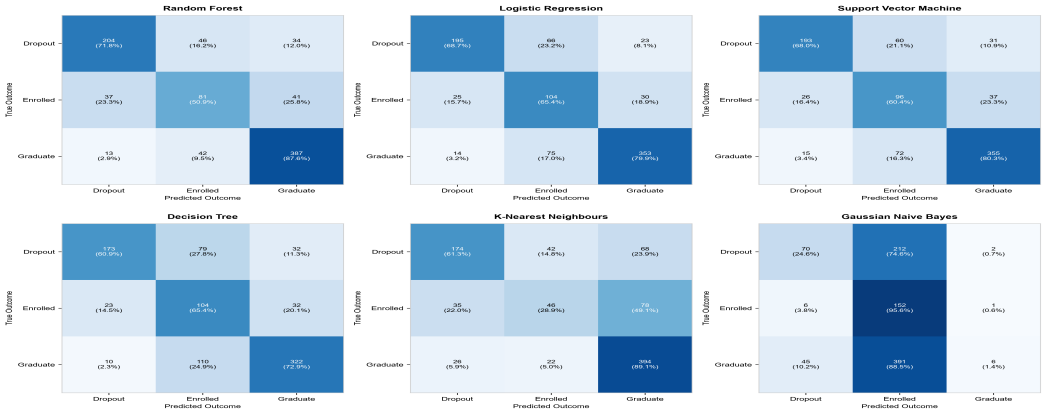


Figure 3: Confusion Matrices for All 6 Classical Classifiers on Held-Out Test Cohort (N=885).

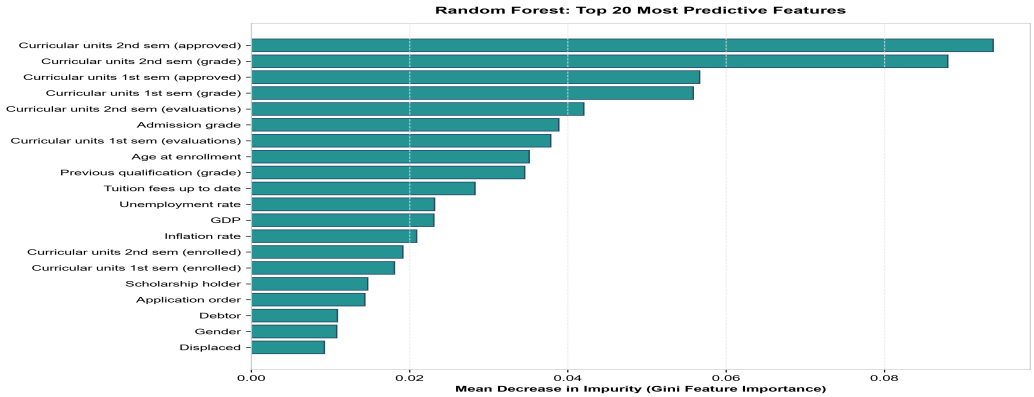


Figure 4: Top 20 Most Predictive Features Ranked by Gini Impurity Reduction (Random Forest).

7.2 Feature Importance Insights: Post-enrollment curricular features (Curricular units 2nd sem approved, 2nd sem grade, 1st sem approved, 1st sem grade) overwhelmingly dominate the model's split decisions. Institutional indicators (Tuition fees up to date, Admission grade) follow closely, while demographic variables contribute minor marginal signal. This reinforces our ethical position that retention is driven primarily by active academic engagement and institutional barriers, not demographic identity.

8. User Interface Architecture & Deployment

8.1 Design Philosophy: Restrained Ergonomics: The GradeGuard interface was developed using Streamlit 1.58, adhering to professional UI/UX standards. To avoid the visual clutter of generic AI dashboards, we utilized a restrained navy (#1B365D), teal (#008080), and off-white palette. Typography prioritizes readability, whitespace, and clear typographic hierarchy, avoiding excessive gradients, neon borders, and frivolous emojis.

8.2 Navigation & Modular Architecture: The application is structured into six dedicated modules:
• *Home (app.py):* System overview, core metrics, temporal scope notices, and probability calibration disclaimers.
• *Predict Student Outcome (pages/1_Predict_Student_Outcome.py):* Interactive 36-feature input form, benchmark profile preloading, CSV batch uploads, probability charts, and supportive guidance.
• *Model Comparison (pages/2_Model_Comparison.py):* Real test metrics, 5-fold CV statistics, confusion matrices, ROC/PR curves, and CSV download links.
• *Dataset Explorer (pages/3_Dataset_Explorer.py):* Interactive distributions, missingness inspection, feature summaries, and cohort data preview.
• *Methodology (pages/4_Methodology.py):* Experimental protocol flowcharts and literature review synthesis table.
• *About & Limitations (pages/5_About_and_Limitations.py):* System boundaries, ethical principles, and execution reproduction commands.

8.3 Supportive Guidance Engine: Rather than outputting sterile classification labels, GradeGuard includes an integrated rule-based decision support engine that generates constructive, supportive recommendations based on observable indicators:
• Low Credit Approval Rate (<60%): Flags credit progression deficits and suggests peer tutoring or study skills coaching.
• Tuition / Debtor Flags: Recommends emergency financial aid counseling, installment payment plans, or student work-study consultations.
• Scholarship Holders at Risk: Triggers proactive check-ins with scholarship retention coordinators.
• Active Enrolled Students: Suggests degree roadmap audits to identify bottleneck prerequisites.
All recommendations include explicit notices that human advisor review is mandatory before any action is initiated.

8.4 Cross-Platform Deployment & Testing: The application is fully container-ready and tested on Windows 11 and Linux, with native support for Streamlit Community Cloud. An automated test suite (pytest) verifies 14 separate unit assertions spanning data loading, preprocessing pipelines, model artifacts, metric integrity, and prediction handlers, executing cleanly in under 3 seconds.

9. Ethical Governance, Limitations, Conclusion & References

9.1 Ethical Governance Principles:

1. *Prohibition of Automated Punitive Actions:* Model outputs must NEVER automate scholarship cancellations, administrative disqualification, or course gating. Automated systems are decision-support prototypes, not arbiters of student futures.
2. *Mandatory Human-in-the-Loop:* Academic advisors, counselors, and faculty mentors must interpret predictions within the qualitative context of individual student lives.
3. *Correlation vs. Causation:* Statistical feature importances do not constitute causal proof. A student with tuition debt does not drop out *because* of the debt variable; underlying family, medical, or socioeconomic pressures must be investigated with empathy.

9.2 Technical & Institutional Limitations:

1. *Single-Institution Origin:* Data originates from the Instituto Politécnico de Portalegre; educational frameworks, grading scales, and support systems differ across borders.
2. *Post-First-Year Feature Dependency:* GradeGuard relies on 1st and 2nd semester academic metrics; it cannot function as an admissions-day early warning tool.
3. *Softmax Heuristics:* Output probabilities represent uncalibrated model scores and must not be treated as actuarial percentages.

9.3 Conclusion: GradeGuard demonstrates that classical machine learning algorithms—when embedded within leak-free pipelines, tuned via stratified cross-validation, and paired with cost-sensitive class weighting—deliver robust, explainable student trajectory predictions (75.93% accuracy, 0.7028 Macro F1, 0.8849 ROC-AUC). By coupling quantitative rigor with human-centered ethical guardrails, the system offers higher education institutions an actionable foundation for equitable student retention.

Scholarly References (Verified DOIs):

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